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Multi-Currency Portfolio Optimization with General Risk Measures

Master Thesis

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Acknowledgments

Abstract

Multi-currency portfolio construction requires managing asset and currency exposures together, yet it remains unclear whether these decisions are best made simultaneously or sequentially as the investable universe grows. This thesis builds on the regularized multi-currency Expected Shortfall framework of Lucescu and Ulrych (2026), in which portfolios are formed by minimizing a risk objective subject to L^1 and L^2 penalties, with currency exposures either optimized jointly with assets or handled through a separate overlay. First we replicate the framework's core strategies, then extend it along two dimensions, the size of the investable universe, and the choice of objective function, implementing both Expected Shortfall and mean-variance formulations of the joint and overlay optimizers.

Across both objectives, the joint optimizer's out-of-sample performance deteriorates relative to the separate overlay as the universe expands. This motivates the central question of the thesis, is this deterioration a structural property of joint optimization, or is it modulated by the choice of objective function? To answer it, we develop a general regularized optimization framework that admits arbitrary objective functions within a single, common formulation. We validate the framework by showing that it exactly reproduces the existing Expected Shortfall and mean-variance optimizers, and then use it to measure how the rate of the joint optimizer's breakdown varies as dimensionality increases.

Sentence on results.

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Chapter 1

Introduction

1.1 Motivation: International Diversification, Currency Risk, and Tail Risk

1.2 Contributions

1.3 Thesis Structure

Chapter 2

Literature Review

Cite the following: International allocation and currency risk management, Currency overlay vs. joint optimization, Downside risk measures and Expected Shortfall, Instability of ES optimization and the role of regularization, Generalized objectives and decision-focused / end-to-end learning.

In this chapter, we review the relevant literature on multi-currency portfolio optimization. We discuss the contributions and methodologies in the field, highlighting how our work builds upon and extends existing research.

The core of our analysis builds on two recent studies that apply regularization techniques to international asset allocation. Burkhardt and Ulrych (2023) introduce a joint optimization approach within a mean-variance framework. By using L1 and L2 regularization, they show that jointly optimizing assets and currencies leads to better out-of-sample performance than the standard industry practice of using a separate currency overlay.

Extending this framework, Lucescu and Ulrych (2026) consider an Expected Shortfall optimization framework to better manage downside tail risk. To fix the numerical instability common in Expected Shortfall minimization, they apply similar regularization methods. Their findings reveal a dimensional trade-off. Joint optimization works best in small asset universes, but the separate currency overlay approach becomes more effective for larger universes due to its built-in dimensionality reduction.

Reproducing and comparing these two baseline frameworks serves as the starting point for the extensions proposed in this thesis.

Chapter 3

Optimization Framework

This chapter develops the optimization framework used throughout the thesis, building up from a model of multi-currency portfolio returns to the risk objectives, the regularization that stabilizes them, and the two formulations through which asset and currency decisions are combined. The derivations follow the methodology presented in [11] and [10], use simple returns, and assume that rebalancing occurs at the end of each investment period. We begin in Section 3.1 with a model-free representation of one-period portfolio returns for an investor who holds assets denominated in multiple currencies and can take currency forward positions for hedging, decomposing the unhedged return of a foreign asset into its local, currency, and cross-currency components, introducing the currency-mapping matrix that links assets to their implicit currency exposures, and expressing the hedged portfolio return in terms of net currency exposures.

With the return model in place, we formulate the optimization problem. Section 3.2 states the classical mean-variance criterion, and Section 3.3 introduces Expected Shortfall as a coherent downside-risk measure, together with the Rockafellar–Uryasev representation that casts its minimization as a tractable linear program. Both are presented as instances of a common mean–risk objective, so that the framework treats the risk measure as a swappable component rather than a fixed choice. Section **??** then introduces the L_1 and L_2 penalties and the leverage constraint used to stabilize the optimization and improve out-of-sample performance.

Finally, Sections **??** and **??** assemble these elements into the two approaches to multi-currency allocation studied throughout the thesis: the separate (currency overlay) formulation, in which asset and currency decisions are made sequentially, and the joint formulation, in which they are optimized simultaneously. The detailed derivations of the resulting quadratic programs are found in Appendix A and Appendix B.

3.1 Model-Free Multi-Currency Portfolio Returns

3.1.1 Unhedged Portfolio Returns

We consider an international investor who allocates their wealth across N assets denominated in up to $M + 1$ currencies. We fix a discrete time index $t \in \{0, 1, 2, \dots\}$ and one investment period from t to $t+1$. Let the investor's *domestic currency* be indexed by 0 and let the remaining M foreign currencies be indexed by $\mathcal{C} \coloneqq \{1, 2, \dots, M\}$.

Let $P_{i,t} > 0$ be the price of asset i at time t in its local currency $c_i \in \mathcal{C} \cup \{0\}$. Define the

corresponding *simple local-currency asset return* over $(t, t{+}1]$ as

$$R_{i,t+1} \coloneqq \frac{P_{i,t+1} - P_{i,t}}{P_{i,t}}.$$

Thus $P_{i,t+1} = P_{i,t}(1 + R_{i,t+1})$.

For each foreign currency $c \in \mathcal{C}$, let $S_{c,t} > 0$ denote the exchange rate at time t, quoted as units of domestic currency per one unit of foreign currency c. The associated *simple currency return* over $(t, t{+}1]$ is

$$e_{c,t+1} \coloneqq \frac{S_{c,t+1} - S_{c,t}}{S_{c,t}}.$$

Equivalently, $S_{c,t+1} = S_{c,t}(1+e_{c,t+1})$. For the domestic currency, we set $S_{0,t} \equiv 1$ and $e_{0,t+1} \equiv 0$.

The domestic-currency value of holding one unit of asset i at time t is $P_{i,t}\, S_{c_i,t}$. Therefore, the *unhedged domestic-currency simple return* of asset i is

$$\begin{aligned} R^u_{i,t+1} &= \frac{P_{i,t+1}S_{c_i,t+1} - P_{i,t}S_{c_i,t}}{P_{i,t}S_{c_i,t}} \\ &= \frac{P_{i,t}(1 + R_{i,t+1})\, S_{c_i,t}(1 + e_{c_i,t+1}) - P_{i,t}S_{c_i,t}}{P_{i,t}S_{c_i,t}} \\ &= R_{i,t+1} + e_{c_i,t+1} + R_{i,t+1}\, e_{c_i,t+1}. \end{aligned} \tag{3.1}$$

Let $\boldsymbol{x}_t \in \mathbb{R}^N$ be the vector of asset portfolio weights chosen at time t and held until $t{+}1$, where $x_{i,t}$ is the fraction of total portfolio value invested in asset i at time t. We impose the standard budget constraint $\mathbf{1}^\mathsf{T}\boldsymbol{x}_t = 1$.

Assuming no rebalancing between times t and $t{+}1$, the domestic-currency portfolio's unhedged return is given by

$$R^u_{\mathcal{P},t+1} = \sum_{i=1}^{N} x_{i,t}\, R^u_{i,t+1}.$$

3.1.2 Hedged Portfolio Returns

To manage currency risk, the investor can take positions in one-period currency forward contracts. For each foreign currency $c \in \mathcal{C}$, let $F_{c,t}$ be the forward rate quoted at time t for settlement at $t{+}1$, in units of domestic currency per unit of foreign currency c. We define the *forward premium* by

$$f_{c,t} \coloneqq \frac{F_{c,t} - S_{c,t}}{S_{c,t}}.$$

The investor hedges by selling foreign currency forward. Let $\phi_{c,t}$ denote the relative notional of the *short* forward position in currency c at time t, expressed as a fraction of total portfolio value, with the convention that $\phi_{c,t} > 0$ corresponds to hedging. Per unit of notional, the short position pays $f_{c,t} - e_{c,t+1}$ in domestic currency over the period. Adding these forward payoffs to the unhedged asset returns, the *hedged portfolio return* is

$$R^h_{\mathcal{P},t+1} = \underbrace{\sum_{i=1}^{N} x_{i,t}\, R^u_{i,t+1}}_{R^u_{\mathcal{P},t+1}} + \sum_{c=1}^{M} \phi_{c,t}\,(f_{c,t} - e_{c,t+1}). \tag{3.2}$$

3.1.3 Matrix Representation and Decomposition

To express the returns in a compact form suitable for optimization, we define the currency incidence matrix $\boldsymbol{C} \in \{0,1\}^{N\times M}$ by

$$C_{i,c} \coloneqq \begin{cases} 1, & \text{if asset } i \text{ is denominated in foreign currency } c, \\ 0, & \text{otherwise.} \end{cases}$$

Each row of $\boldsymbol{C}$ has at most one nonzero entry, and the rows corresponding to domestic assets are identically zero. This allows us to define the portfolio's *implicit foreign-currency exposure* as

$$\boldsymbol{w}_t \coloneqq \boldsymbol{C}^\mathsf{T}\boldsymbol{x}_t \in \mathbb{R}^M, \qquad w_{c,t} = \sum_{i:\, c_i=c} x_{i,t},$$

such that $w_{c,t}$ is the total portfolio weight invested in assets denominated in currency c. The implicit domestic exposure follows residually as $w_{0,t} = 1 - \mathbf{1}_M^\mathsf{T}\boldsymbol{w}_t$, since the asset weights sum to one.

We express returns in vector form $\boldsymbol{R}_{t+1} \in \mathbb{R}^N$, $\boldsymbol{e}_{t+1} \in \mathbb{R}^M$, and the forward premiums as $\boldsymbol{f}_t \in \mathbb{R}^M$. The product $\boldsymbol{C}\boldsymbol{e}_{t+1} \in \mathbb{R}^N$ aligns currency returns to assets, with ith component $e_{c_i,t+1}$ for foreign assets and 0 for domestic ones. Using (3.1), the vector of *unhedged domestic returns* across assets can be written as

$$\boldsymbol{R}^u_{t+1} = \boldsymbol{R}_{t+1} + \boldsymbol{C}\boldsymbol{e}_{t+1} + (\boldsymbol{R}_{t+1} \odot \boldsymbol{C}\boldsymbol{e}_{t+1}),$$

where $\odot$ denotes the Hadamard product.

Expressing the forward positions in vector form as $\boldsymbol{\phi}_t \in \mathbb{R}^M$, the hedged portfolio return (3.2) can be written as

$$R^h_{\mathcal{P},t+1} = \boldsymbol{x}_t^\mathsf{T}\boldsymbol{R}^u_{t+1} + \boldsymbol{\phi}_t^\mathsf{T}(\boldsymbol{f}_t - \boldsymbol{e}_{t+1}). \tag{3.3}$$

Setting $\boldsymbol{\phi}_t = \boldsymbol{w}_t = \boldsymbol{C}^\mathsf{T}\boldsymbol{x}_t$ neutralizes the implicit currency exposure. Substituting this together with the unhedged return vector into (3.3), the *fully hedged portfolio return* is

$$\begin{aligned} R^{fh}_{\mathcal{P},t+1} &= \boldsymbol{x}_t^\mathsf{T}\boldsymbol{R}^u_{t+1} + (\boldsymbol{C}^\mathsf{T}\boldsymbol{x}_t)^\mathsf{T}(\boldsymbol{f}_t - \boldsymbol{e}_{t+1}) \\ &= \boldsymbol{x}_t^\mathsf{T}\Big[\boldsymbol{R}_{t+1} + \boldsymbol{C}\boldsymbol{e}_{t+1} + (\boldsymbol{R}_{t+1} \odot \boldsymbol{C}\boldsymbol{e}_{t+1})\Big] + \boldsymbol{x}_t^\mathsf{T}\boldsymbol{C}(\boldsymbol{f}_t - \boldsymbol{e}_{t+1}) \\ &= \boldsymbol{x}_t^\mathsf{T}\Big[\boldsymbol{R}_{t+1} + \boldsymbol{C}\boldsymbol{e}_{t+1} + (\boldsymbol{R}_{t+1} \odot \boldsymbol{C}\boldsymbol{e}_{t+1}) + \boldsymbol{C}\boldsymbol{f}_t - \boldsymbol{C}\boldsymbol{e}_{t+1}\Big] \\ &= \boldsymbol{x}_t^\mathsf{T}\Big[\boldsymbol{R}_{t+1} + \boldsymbol{C}\boldsymbol{f}_t + (\boldsymbol{R}_{t+1} \odot \boldsymbol{C}\boldsymbol{e}_{t+1})\Big] \\ &= \boldsymbol{x}_t^\mathsf{T}\boldsymbol{R}^{fh}_{t+1}. \end{aligned}$$

The first-order currency terms $\pm\boldsymbol{x}_t^\mathsf{T}\boldsymbol{C}\boldsymbol{e}_{t+1}$ cancel, leaving the *fully hedged asset return vector*

$$\boldsymbol{R}^{fh}_{t+1} = \boldsymbol{R}_{t+1} + \boldsymbol{C}\boldsymbol{f}_t + (\boldsymbol{R}_{t+1} \odot \boldsymbol{C}\boldsymbol{e}_{t+1}).$$

More generally, define the *net currency exposure* $\boldsymbol{\psi}_t \coloneqq \boldsymbol{w}_t - \boldsymbol{\phi}_t$. Imposing $\sum_{c=0}^M \phi_{c,t} = 1$ gives $\sum_{c=0}^M \psi_{c,t} = 0$, so the currency overlay is a zero-investment portfolio. Substituting $\boldsymbol{\phi}_t = \boldsymbol{w}_t - \boldsymbol{\psi}_t$ into (3.3) expresses the hedged return in terms of net exposures,

$$\begin{aligned} R^h_{\mathcal{P},t+1} &= \boldsymbol{x}_t^\mathsf{T}\boldsymbol{R}^u_{t+1} + (\boldsymbol{w}_t - \boldsymbol{\psi}_t)^\mathsf{T}(\boldsymbol{f}_t - \boldsymbol{e}_{t+1}) \\ &= \boldsymbol{x}_t^\mathsf{T}\boldsymbol{R}^u_{t+1} + \boldsymbol{w}_t^\mathsf{T}(\boldsymbol{f}_t - \boldsymbol{e}_{t+1}) + \boldsymbol{\psi}_t^\mathsf{T}(\boldsymbol{e}_{t+1} - \boldsymbol{f}_t) \\ &= \boldsymbol{x}_t^\mathsf{T}\boldsymbol{R}^{fh}_{t+1} + \boldsymbol{\psi}_t^\mathsf{T}(\boldsymbol{e}_{t+1} - \boldsymbol{f}_t), \end{aligned} \tag{3.4}$$

which we adopt throughout the following portfolio optimization formulations, as it makes the asset weights $\boldsymbol{x}_t$ and the net currency exposures $\boldsymbol{\psi}_t$ the explicit variables of the optimization.

3.2 Mean-Variance Optimization Formulation

3.2.1 The Mean-Variance Criterion

The mean-variance criterion of [6] measures portfolio risk by the variance of the portfolio return and selects weights that balance this risk against expected return. Using the hedged portfolio return $R^h_{\mathcal{P},t+1}$ defined in (3.4) and a relative risk-aversion parameter $\gamma > 0$, the investor solves

$$\min \ \frac{\gamma}{2}\operatorname{Var}\Big(R^h_{\mathcal{P},t+1}\Big) - \mathbb{E}\Big[R^h_{\mathcal{P},t+1}\Big], \qquad \text{s.t.} \quad \mathbf{1}_N^\top \boldsymbol{x}_t = 1. \tag{3.5}$$

Relative risk-aversion γ governs the trade-off between variance and expected return, larger values place more weight on risk reduction, and in the limit $\gamma \to \infty$ the criterion reduces to pure variance minimization.

The hedged portfolio return (3.4) is linear in the asset weights $\boldsymbol{x}_t$ and the net currency exposures $\boldsymbol{\psi}_t$. Consequently $\mathbb{E}[R^h_{\mathcal{P},t+1}]$ is linear and $\operatorname{Var}(R^h_{\mathcal{P},t+1})$ is a quadratic form in these decision variables, so that, once the expected-return vector $\boldsymbol{\mu}$ and the covariance matrix. $\boldsymbol{\Sigma}$ are estimated, (3.5) can be formulated as a convex quadratic program. Whether the asset and currency weights are chosen sequentially or in a single step determines the precise decision vector and the accompanying constraints. These two formulations are developed in Sections 3.2.3 and 3.2.4.

While analytically convenient, variance is not a coherent risk measure in the sense of [1], who characterize a coherent measure by monotonicity, subadditivity, positive homogeneity, and translation invariance. For example, variance violates the monotonicity property, a portfolio whose return dominates another need not have lower variance, so variance can rank a uniformly preferable position as the riskier one.[1] Equally relevant for international portfolios exposed to regime shifts and market stress, variance penalizes upside and downside deviations symmetrically and therefore does not isolate the large losses that most concern investors. These shortcomings motivate the coherent, downside-focused alternative developed in Section 3.3, where we consider Expected Shortfall as the risk measure in the portfolio optimization problem.

3.2.2 Regularized Mean-Variance Problem

The mean-variance criterion in (3.5) relies on the expected-return vector $\boldsymbol{\mu}$ and the covariance matrix $\boldsymbol{\Sigma}$, which are unknown and must be estimated from historical data. Replacing them with their sample counterparts $\hat{\boldsymbol{\mu}}$ and $\hat{\boldsymbol{\Sigma}}$ yields the plug-in solution, which is known to perform poorly out of sample: the optimizer is highly sensitive to estimation error and tends to produce extreme, unstable, and concentrated weights [**?** **?**]. The problem intensifies as the number of assets and currencies grows relative to the sample size, and is therefore especially pronounced in the joint formulation, where the asset–currency dependencies must additionally be estimated.

To control this estimation error, we regularize the objective, deliberately introducing bias in exchange for a reduction in variance that improves out-of-sample performance [**?** **?**]. Following [11], we employ three regularization devices: an L_1 penalty, an L_2 penalty, and a leverage constraint. Applied to the asset weights $\boldsymbol{x}_t$ and the net currency exposures $\boldsymbol{\psi}_t$, the regularized mean-variance problem reads

$$\min \ \frac{\gamma}{2}\operatorname{Var}\Big(R^h_{\mathcal{P},t+1}\Big) - \mathbb{E}\Big[R^h_{\mathcal{P},t+1}\Big] + \lambda_1^a\|\boldsymbol{x}_t\|_1 + \lambda_2^a\|\boldsymbol{x}_t\|_2^2 + \lambda_1^c\|\boldsymbol{\psi}_t\|_1 + \lambda_2^c\|\boldsymbol{\psi}_t\|_2^2, \quad \text{s.t.} \quad \mathbf{1}_N^\top \boldsymbol{x}_t = 1, \ \ \|\boldsymbol{x}_t\|_1 \le \ell, \tag{3.6}$$

[1] Consider $X \sim U(0,1)$ and $Y \sim U(1,3)$, then $X \le Y$ a.s., but $\operatorname{Var}(X) < \operatorname{Var}(Y)$.

where the moments are evaluated at the sample estimates $\hat{\boldsymbol{\mu}}$ and $\hat{\boldsymbol{\Sigma}}$. In the separate formulation these penalties enter the asset and currency steps individually, whereas in the joint formulation they enter a single objective; both are developed in Sections 3.2.3 and 3.2.4.

The L_1 penalty, $\|\boldsymbol{x}_t\|_1 = \sum_i |x_{i,t}|$, induces sparsity: because its constraint region is non-smooth, the optimum is typically attained at a vertex where several weights are exactly zero [**?**]. Sparse portfolios are practically appealing, as they hold fewer active positions and thereby incur lower transaction costs, and the penalty admits a direct interpretation as a proxy for proportional trading costs [**?**]. It additionally limits short-selling and reduces the optimizer's sensitivity to collinearity among the assets, improving numerical stability [**?** **?**].

The L_2 penalty, $\|\boldsymbol{x}_t\|_2^2 = \sum_i x_{i,t}^2$, induces smooth shrinkage: rather than zeroing weights, it pulls them toward a more uniform allocation, damping the large offsetting long–short positions that drive mean-variance instability. In the mean-variance setting this ridge penalty has a direct statistical interpretation: [**?**] shows that constraining the L_2 norm of the weights is equivalent to shrinking the covariance matrix toward the identity, linking it to the covariance-shrinkage estimators of [**?**]. The two penalties are complementary, and their combination is the elastic net.

Estimation error can also be attacked directly at the input level by shrinking the parameter estimates themselves. The sample mean $\hat{\boldsymbol{\mu}}$ is notoriously unreliable and is the dominant source of mean-variance instability [**?** **?**]. We therefore replace it with the Bayes–Stein estimator of [**?**], which shrinks each asset's sample mean toward the common grand mean by an amount that grows with estimation uncertainty, limiting the influence of extreme return estimates on the optimal weights. This shrinkage is analogous to the L_2 penalty on the covariance side: [**?**] shows that the ridge penalty is equivalent to shrinking $\hat{\boldsymbol{\Sigma}}$ toward the identity, the weight-space counterpart of the covariance-shrinkage estimators of [**?**].

Finally, the leverage constraint $\|\boldsymbol{x}_t\|_1 \leq \ell$, with $\ell \geq 1$, bounds the gross asset exposure. It guarantees that the problem remains bounded even when the penalties are absent, since the unregularized mean-variance objective can otherwise exploit near-singular directions in $\hat{\boldsymbol{\Sigma}}$ to take arbitrarily large offsetting positions, and it provides practically relevant control over leverage [**?**]. The choice $\ell = 1$ rules out leverage entirely and coincides with a hard L_1 constraint.

The asset and currency penalties are governed by separate strengths, λ_1^a, λ_2^a for the assets and λ_1^c, λ_2^c for the currencies, rather than a single shared parameter. This distinction is economically motivated: only the asset weights are constrained to sum to one, transaction costs differ between the two markets, and, while sparsity is desirable for asset selection, driving currency exposures to zero would undermine the very currency risk management the framework is designed to provide. Treating the two sets of penalties separately therefore allows assets and currencies to be regularized to different degrees.

All three devices preserve tractability: the L_1 terms are linearized through auxiliary variables that capture the positive and negative parts of the weights, and the L_2 terms retain a quadratic structure, so that the regularized problem remains a quadratic program with linear constraints. A further device specific to the multi-currency setting, a bound on the net currency exposure, likewise acts as a form of shrinkage; because it couples the asset and currency decisions, we introduce it directly in the formulations below. The explicit separate and joint programs, together with the augmented-vector derivations, are given in Sections 3.2.3 and 3.2.4 and in Appendix B.

3.2.3 Currency Overlay Optimization

3.2.4 Joint Optimization

3.3 Expected Shortfall Optimization Formulation

3.3.1 Expected Shortfall and the Rockafellar–Uryasev Representation

3.3.2 Regularized Expected Shortfall Problem

3.3.3 Currency Overlay Optimization

3.3.4 Joint Optimization

Chapter 4

Data and Implementation

4.1 Data and Summary Statistics

Our analysis relies on a comprehensive dataset of international equities, foreign exchange spot rates, and forward exchange rates sourced from Refinitiv Datastream. The dataset covers a period from June 1990 to June 2023, capturing various market regimes, including the dot-com bubble burst, the 2008 Global Financial Crisis, the European debt crisis, and the COVID-19 pandemic.

We construct our investment universe from the 14 largest developed equity markets: Australia, Canada, France, Germany, Hong Kong, Italy, Japan, the Netherlands, Singapore, Spain, Sweden, Switzerland, the United Kingdom, and the United States. To construct the local equity portfolios, we select the largest stocks in each market based on their market capitalization at the beginning of each year. We consider two distinct cross-sections of stocks to analyze the impact of the investment universe size on portfolio performance, a small universe consisting of the top 10 stocks per market (140 stocks in total) and a large universe consisting of the top 50 stocks per market (700 stocks in total).

For each selected stock, we retrieve daily total return indices, which account for dividend reinvestments, and compute the corresponding local currency returns. The US dollar serves as the base currency for our analysis, we calculate the fully hedged returns for each stock following the model-free return representation in chapter **??**.

4.2 Modular Backtesting Framework

This chapter describes the implementation of the walk-forward backtesting framework used throughout the thesis. The code is organised around a small number of clearly separated components: a data handler that prepares the investment universe and return series, a backtester that orchestrates the rolling evaluation, and a family of optimizers that implement the portfolio construction problem.[1] This modular structure keeps the experimental setup reproducible while allowing different objective functions and portfolio architectures to share the same evaluation pipeline.

The figure highlights the main design principle of the implementation: the backtester is intentionally kept generic, while the portfolio logic lives in interchangeable optimizer classes. This makes it possible to compare mean-variance and expected shortfall formulations, asset-

[1]The full codebase for this project is publicly available: https://github.com/anejrozman/Multi-Currency-Portfolio-Optimization-with-General-Risk-Measures.

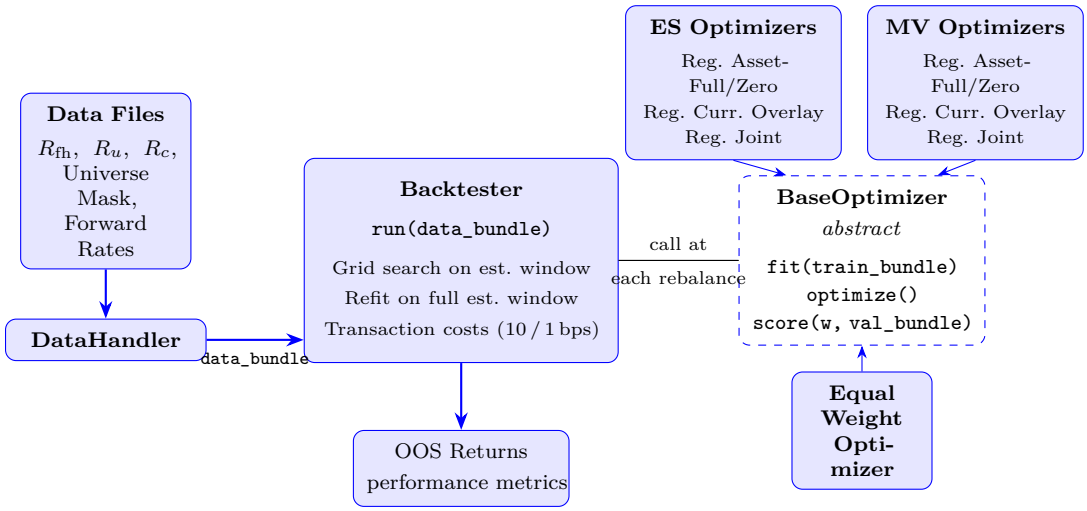


Figure 4.1: Schematic of the backtesting framework.

only and joint asset–currency strategies, and regularized and unregularized variants under the same walk-forward protocol.

4.2.1 Backtesting Workflow and Optimizer Structure

The framework is built around a strict separation of concerns. The `DataHandler` prepares the inputs for the backtest by loading precomputed return series, mapping each asset to its reporting currency, and building a daily universe mask from the investment universe file. The backtester receives these objects as a single `data_bundle` and does not need to know how they were generated.

The `Backtester` is responsible for the evaluation protocol. It determines rebalancing dates, constructs estimation and validation windows, performs hyperparameter selection when a grid is supplied, and records the resulting out-of-sample returns. All portfolio construction decisions are delegated to subclasses of `BaseOptimizer`, which all implement `fit(train_bundle)`, `optimize()`, and `score(weights, val_bundle)`.

The backtest follows a standard walk-forward procedure. At each rebalance date, the framework forms a rolling estimation window and a separate validation window, evaluates candidate hyperparameters, refits the best configuration on the full estimation sample, and then carries the portfolio into the next holding period. Rebalancing can be daily, weekly, monthly, quarterly, semi-annual, or annual. Before optimization, the code filters the investable universe to currently eligible assets and checks that each selected asset has a complete return history, which keeps missing-data handling outside the optimization problem. Window lengths are stored centrally in `backtest_config.py`; the default setup uses a ten-year estimation window and a one-year validation window.

4.2.2 Optimization, Costs, and Outputs

Two optimizer families are implemented: expected shortfall and mean-variance. Expected shortfall is formulated as a convex program with auxiliary variables, optional ℓ_1 and ℓ_2 regularization, and an ℓ_1-norm leverage cap. Mean-variance uses Bayes–Stein shrinkage for expected returns and a lightly regularized covariance matrix. Both families are available in asset-only and joint asset–currency variants. The asset-only strategies use either hedged returns (*assets-full*) or unhedged returns (*assets-zero*), while the joint strategies solve for asset

weights and currency hedge weights together. The equal-weight benchmark follows the same interface but returns a uniform allocation.

All substantive optimizers are solved with CVXPY. When available, Gurobi is used as the default solver with tight tolerances; otherwise, the code falls back to another compatible solver. Transaction costs are applied at each rebalance using separate assumptions for equities and FX, namely 10 basis points and 1 basis point. Costs are computed with a log-linear turnover approximation, and the backtester tracks drifted weights between rebalances so that turnover reflects actual portfolio evolution. The framework outputs daily out-of-sample returns and summary statistics including annualized volatility, maximum drawdown, Sharpe ratio, Sortino ratio, and an expected shortfall ratio.

4.3 Numerical Implementation of Optimization Problems

4.4 Evaluation Protocol and Performance Metrics

Chapter 5

Empirical Analysis and Baseline Results

In this chapter, we present an empirical analysis of the multi-currency portfolio optimization problem using historical data. We apply the model-free return representation derived in chapter **??** to compute realized returns for the portfolio strategies derived in section **??**. We evalutate the performance of the various strategies on two investment universes, a large investment universe consisting of 50 biggest stocks in the 14 largest developed markets and a small investment universe consisting of 10 biggest stocks in the same markets. We compare the performance of the optimized portfolios to a benchmark equally weighted portfolio.

5.1 Backtest Results of Expected Shortfall Optimization Approaches

We can see in table 5.1 that the performance of the expected shortfall optimization approaches varies across different universe sizes. In the small universe, the optimized portfolios generally outperform the equal weight benchmark in terms of risk-adjusted returns, as measured by the Sharpe Ratio and Sortino Ratio. However, as we increase the universe size to medium and large, the performance of the optimized portfolios relative to the equal weight benchmark deteriorates, with some strategies even underperforming the benchmark in terms of risk-adjusted returns. This suggests that the benefits of optimization may diminish as the investment universe expands, potentially due to increased estimation error and model uncertainty in larger universes.

5.2 Backtest Results of Mean-Variance Optimization Approaches

5.3 The Effects of Universe Size on Portfolio Performance

Panel A: Small Investment Universe (top 10 stocks per economy)							
Metric	**Equal Weight**	**Assets-Full**	**Assets-Zero**	**Overlay**	**Overlay-Reg**	**Joint**	**Joint-Reg**
ES (%)							
Vol (%)							
MD (%)							
SR							
SOR							
ESR							
Turnover^a							
Turnover^c							
Panel B: Medium Investment Universe (top 50 stocks per economy)							
Metric	**Equal Weight**	**Assets-Full**	**Assets-Zero**	**Overlay**	**Overlay-Reg**	**Joint**	**Joint-Reg**
ES (%)							
Vol (%)							
MD (%)							
SR							
SOR							
ESR							
Turnover^a							
Turnover^c							
Panel C: Large Investment Universe (top 100 stocks per economy)							
Metric	**Equal Weight**	**Assets-Full**	**Assets-Zero**	**Overlay**	**Overlay-Reg**	**Joint**	**Joint-Reg**
ES (%)							
Vol (%)							
MD (%)							
SR							
SOR							
ESR							
Turnover^a							
Turnover^c							

Table 5.1: Performance metrics for Expected Shortfall optimization approaches on increasing universe sizes

Table 5.1 reports out-of-sample performance metrics for the international portfolio optimization under expected shortfall, with portfolios rebalanced quarterly. The realized out-of-sample statistics shown are Expected Shortfall (ES), volatility (Vol), maximum drawdown (MD), the Sharpe Ratio (SR), the Sortino Ratio (SOR), the Expected Shortfall Ratio (ESR), and average asset and currency turnover, denoted Turnover^a and Turnover^c, respectively. Each panel corresponds to a different investment universe: Panel A covers the small universe, made up of the 10 largest companies by market capitalization in each economy; Panel B the medium universe, comprising the 50 largest companies per economy; and Panel C the large universe, consisting of the 100 largest companies per economy. The sample period runs from June 1990 to June 2023, and the first out-of-sample portfolio is formed in June 2000. Details on the optimization approaches and performance metrics are provided in section **??**.

Panel A: Small Investment Universe (top 10 stocks per economy)							
Metric	**Equal Weight**	**Assets-Full**	**Assets-Zero**	**Overlay**	**Overlay-Reg**	**Joint**	**Joint-Reg**
ES (%)							
Vol (%)							
MD (%)							
SR							
SOR							
ESR							
Turnover^a							
Turnover^c							
Panel B: Medium Investment Universe (top 50 stocks per economy)							
Metric	**Equal Weight**	**Assets-Full**	**Assets-Zero**	**Overlay**	**Overlay-Reg**	**Joint**	**Joint-Reg**
ES (%)							
Vol (%)							
MD (%)							
SR							
SOR							
ESR							
Turnover^a							
Turnover^c							
Panel C: Large Investment Universe (top 100 stocks per economy)							
Metric	**Equal Weight**	**Assets-Full**	**Assets-Zero**	**Overlay**	**Overlay-Reg**	**Joint**	**Joint-Reg**
ES (%)							
Vol (%)							
MD (%)							
SR							
SOR							
ESR							
Turnover^a							
Turnover^c							

Table 5.2: Performance metrics for mean-variance optimization approaches on increasing universe sizes

Table 5.2 reports out-of-sample performance metrics for the international portfolio optimization under mean-variance, with portfolios rebalanced quarterly. The realized out-of-sample statistics shown are Expected Shortfall (ES), volatility (Vol), maximum drawdown (MD), the Sharpe Ratio (SR), the Sortino Ratio (SOR), the Expected Shortfall Ratio (ESR), and average asset and currency turnover, denoted Turnover^a and Turnover^c, respectively. Each panel corresponds to a different investment universe: Panel A covers the small universe, made up of the 10 largest companies by market capitalization in each economy; Panel B the medium universe, comprising the 50 largest companies per economy; and Panel C the large universe, consisting of the 100 largest companies per economy. The sample period runs from June 1990 to June 2023, and the first out-of-sample portfolio is formed in June 2000. Details on the optimization approaches and performance metrics are provided in section **??**.

Chapter 6

Convex Risk Measures

Chapter 7

End To End Portfolio Optimization

Chapter 8

Conclusion

8.1 Findings

8.2 Limitations and Future Research

Appendices

Appendix A

Optimization Derivation: Expected Shortfall

In this appendix, we derive the optimization problems used in the main text, building up from the Expected Shortfall objective to the implementable quadratic programs solved in the empirical analysis. In Section A.1 we follow [8] to derive the equivalence between direct minimization of Expected Shortfall and an auxiliary formulation suitable for linear programming. Following [10], in Section A.2 we embed this equivalence in a mean-ES objective, by replacing the conditional expectation with its empirical counterpart, and linearizing the terms to assemble the base unregularized portfolio optimization problem as a linear program. In Section A.3 we introduce three regularization techniques, an L_1 penalty, an L_2 penalty, and a leverage constraint, giving us the base regularized quadratic program for portfolio optimization. In Section A.4 we extend the framework to the multi-currency setting under the separate (currency overlay) approach, deriving the asset and the currency quadratic programs. Finally, in Section A.5 we derive the joint quadratic program formulation, in which asset and currency weights are optimized simultaneously through a single Expected Shortfall term that captures the combined return distribution.

A.1 Derivation of the Auxiliary Expected Shortfall Formulation

We derive the equivalence between the direct minimization of Expected Shortfall and the auxiliary formulation proposed by Rockafellar and Uryasev [8] that allows for formulating the portfolio optimization as a linear program.

We assume that the loss distribution is continuous, meaning the loss cumulative distribution function has no jumps and the probability density function $f(l)$ exists. Under these assumptions, Value-at-Risk (VaR) is uniquely defined [7].

Define the loss random variable at time $t+1$, given portfolio weights x_t, as $L(x_t) = -x_t^T R_{t+1}$. Let $f(l)$ denote the probability density function of the loss $L(x_t)$. The VaR at confidence level β is defined as the β-quantile of the loss distribution

$$\mathrm{VaR}_\beta(x_t) = \min\{\alpha : \mathbb{P}(L(x_t) \leq \alpha) \geq \beta\}.$$

The Expected Shortfall at confidence level β is defined as the conditional expectation of losses exceeding VaR_β

$$\mathrm{ES}_\beta(x_t) = \mathbb{E}[L(x_t) \mid L(x_t) \geq \mathrm{VaR}_\beta(x_t)] = \frac{1}{1-\beta} \int_{\mathrm{VaR}_\beta(x_t)}^{\infty} l f(l)\, dl.$$

We now consider the auxiliary function F_β defined as

$$F_\beta(x_t, \alpha) = \alpha + \frac{1}{1-\beta}\mathbb{E}_t[(L(x_t) - \alpha)^+].$$

We can rewrite the expectation term using the integral form

$$\mathbb{E}_t[(L(x_t) - \alpha)^+] = \int_{-\infty}^{\infty} (l-\alpha)^+ f(l)\, dl = \int_{\alpha}^{\infty} (l-\alpha) f(l)\, dl.$$

Before minimizing $F_\beta(x_t, \alpha)$ with respect to α, we verify convexity. The second derivative of F_β with respect to α is

$$\begin{aligned}\frac{\partial F_\beta}{\partial \alpha} &= 1 - \frac{1}{1-\beta}\int_{\alpha}^{\infty} f(l)\, dl \\ \frac{\partial^2 F_\beta}{\partial \alpha^2} &= \frac{1}{1-\beta} f(\alpha).\end{aligned}$$

Since the probability density function $f(\alpha) \geq 0$ and $\beta < 1$, the second derivative is non-negative, $\frac{\partial^2 F_\beta}{\partial \alpha^2} \geq 0$. Thus, $F_\beta(x_t, \alpha)$ is convex in α, ensuring that the first-order condition is necessary and sufficient for a global minimum.

Differentiation of $F_\beta(x_t, \alpha)$ with respect to α yields the first-order condition for optimality. Using Leibniz's rule we have

$$\begin{aligned}\frac{\partial F_\beta}{\partial \alpha} &= 1 + \frac{1}{1-\beta}\frac{\partial}{\partial \alpha}\int_{\alpha}^{\infty} (l-\alpha) f(l)\, dl \\ &= 1 + \frac{1}{1-\beta}\left[-f(\alpha)\cdot 0 + \int_{\alpha}^{\infty} (-1) f(l)\, dl\right] \\ &= 1 - \frac{1}{1-\beta}\int_{\alpha}^{\infty} f(l)\, dl \\ &= 1 - \frac{1}{1-\beta}\mathbb{P}(L(x_t) \geq \alpha).\end{aligned}$$

Setting the derivative to zero to find the optimal α^*

$$\begin{aligned}1 - \frac{1}{1-\beta}\mathbb{P}(L(x_t) \geq \alpha^*) &= 0 \\ \mathbb{P}(L(x_t) \geq \alpha^*) &= 1-\beta \\ \mathbb{P}(L(x_t) \leq \alpha^*) &= \beta.\end{aligned}$$

This implies that the optimal α^* that minimizes the auxiliary function for a fixed x_t is exactly the Value-at-Risk at level β, so $\alpha^* = \mathrm{VaR}_\beta(x_t)$.

Substituting $\alpha^* = \mathrm{VaR}_\beta(x_t)$ back into $F_\beta(x_t, \alpha)$ we obtain

$$\begin{aligned}F_\beta(x_t, \mathrm{VaR}_\beta(x_t)) &= \mathrm{VaR}_\beta(x_t) + \frac{1}{1-\beta}\int_{\mathrm{VaR}_\beta(x_t)}^{\infty} (l - \mathrm{VaR}_\beta(x_t)) f(l)\, dl \\ &= \mathrm{VaR}_\beta(x_t) + \frac{1}{1-\beta}\left(\int_{\mathrm{VaR}_\beta(x_t)}^{\infty} l f(l)\, dl - \mathrm{VaR}_\beta(x_t)\int_{\mathrm{VaR}_\beta(x_t)}^{\infty} f(l)\, dl\right).\end{aligned}$$

Recognizing that $\int_{\mathrm{VaR}_\beta(x_t)}^{\infty} f(l)\,dl = \mathbb{P}(L(x_t) \geq \mathrm{VaR}_\beta(x_t)) = 1-\beta$, we have

$$\begin{aligned} F_\beta(x_t, \mathrm{VaR}_\beta(x_t)) &= \mathrm{VaR}_\beta(x_t) + \frac{1}{1-\beta}\int_{\mathrm{VaR}_\beta(x_t)}^{\infty} l f(l)\,dl - \frac{1}{1-\beta}\mathrm{VaR}_\beta(x_t)(1-\beta) \\ &= \mathrm{VaR}_\beta(x_t) + \mathrm{ES}_\beta(x_t) - \mathrm{VaR}_\beta(x_t) \\ &= \mathrm{ES}_\beta(x_t). \end{aligned}$$

Thus, minimizing $\mathrm{ES}_\beta(x_t)$ with respect to x_t is equivalent to minimizing $F_\beta(x_t, \alpha)$ jointly with respect to x_t and α

$$\min_{x_t} \mathrm{ES}_\beta(x_t) = \min_{x_t, \alpha} \left(\alpha + \frac{1}{1-\beta}\mathbb{E}_t[(-x_t^T R_{t+1} - \alpha)^+] \right). \tag{A.1}$$

Finally, we observe that $F_\beta(x_t, \alpha)$ is jointly convex in (x_t, α). The term α is linear (hence convex). The inner map $(x_t, \alpha) \mapsto (-x_t^\top R_{t+1} - \alpha)$ is affine. The outer function $z \mapsto z^+$ is convex and nondecreasing. The composition of a convex function with an affine map is convex [2]. Thus $(-x_t^\top R_{t+1} - \alpha)^+$ is jointly convex in (x_t, α). Taking the expectation preserves convexity.

Since the objective function is convex, any local minimum is global, ensuring that optimization algorithms will converge efficiently to the optimal solution.

A.2 Linear Program Representation of Portfolio Optimization

The equivalence in (A.1) reduces Expected Shortfall minimization to the minimization of the jointly convex auxiliary function $F_\beta(x_t, \alpha)$

$$\min_{x_t} \mathrm{ES}_\beta(x_t) = \min_{x_t, \alpha} \left(\alpha + \frac{1}{1-\beta}\mathbb{E}_t[(-x_t^\top R_{t+1} - \alpha)^+] \right).$$

This formulation is not yet appropriate for portfolio optimization, as it only optimizes for downside risk, without taking into account the expected return of the portfolio. It also depends on the unknown expectation $\mathbb{E}_t[\cdot]$. We address this by incorporating the expected return into the optimization objective and replacing the expectation with its empirical estimator.

Following [4] and [10], we trade off expected return against downside risk through a relative risk-aversion coefficient $\gamma > 0$

$$x_t^* = \arg\min_{x_t} \left\{ -\mu_t^\top x_t + \frac{\gamma}{2}\mathrm{ES}_\beta(x_t) \right\}, \quad \text{s.t. } x_t^\top \mathbf{1}_N = 1, \tag{A.2}$$

where μ_t denotes the asset return mean estimator and the budget constraint $x_t^\top \mathbf{1}_N = 1$ enforces full investment. Substituting the auxiliary representation (A.1) into (A.2) gives

$$x_t^*, \alpha^* = \arg\min_{x_t, \alpha} \left\{ -\mu_t^\top x_t + \frac{\gamma}{2}\alpha + \frac{\gamma}{2(1-\beta)}\mathbb{E}_t[(-x_t^\top R_{t+1} - \alpha)^+] \right\}.$$

Let $y_1, \ldots, y_q \in \mathbb{R}^N$ denote q historical realizations of the joint return vector R_{t+1}. We approximate the conditional expectation by the empirical average over this sample

$$\mathbb{E}_t[(-x_t^\top R_{t+1} - \alpha)^+] \approx \frac{1}{q}\sum_{k=1}^{q}(-x_t^\top y_k - \alpha)^+,$$

so that the portfolio optimization problem becomes

$$x_t^*, \alpha^* = \arg\min_{x_t, \alpha} \left\{ -\mu_t^\top x_t + \frac{\gamma}{2}\alpha + \frac{\gamma}{2q(1-\beta)} \sum_{k=1}^{q} (-x_t^\top y_k - \alpha)^+ \right\}, \quad \text{s.t. } x_t^\top \mathbf{1}_N = 1. \tag{A.3}$$

The summand $(-x_t^\top y_k - \alpha)^+$ is convex but non-linear. We linearize it by introducing, for each observation k, an auxiliary variable $u_k \geq 0$ subject to

$$u_k \geq -x_t^\top y_k - \alpha \qquad \Longleftrightarrow \qquad x_t^\top y_k + \alpha + u_k \geq 0. \tag{A.4}$$

Since the objective in (A.3) is strictly increasing in each u_k (its coefficient $\frac{\gamma}{2q(1-\beta)} > 0$) and minimized, every u_k is driven to its lower bound at the optimum, so $u_k^* = (-x_t^\top y_k - \alpha)^+$. The slack constraints therefore reproduce the term $(-x_t^\top y_k - \alpha)^+$ while keeping all relations linear.

By collecting the decision variables in the vector

$$\tilde{x}_t = \Big[x_{1,t}, \ldots, x_{N,t}, \quad \alpha, \quad u_1, \ldots, u_q \Big]^\top = \Big[x_t^\top, \quad \alpha, \quad u^\top \Big]^\top \in \mathbb{R}^{(N+q+1)},$$

and constants in the vector

$$c = \left[-\mu_t^\top, \quad \frac{\gamma}{2}, \quad \frac{\gamma}{2q(1-\beta)} \mathbf{1}_q^\top \right]^\top \in \mathbb{R}^{(N+q+1)},$$

the portfolio optimization problem (A.3) can be written as a linear program

$$\begin{aligned} \tilde{x}_t^* &= \arg\min_{\tilde{x}_t} \quad c^\top \tilde{x}_t \\ \text{s.t.} \quad D\tilde{x}_t &= \Big[\mathbf{1}_N^\top \quad 0_{q+1}^\top \Big] \tilde{x}_t = E = 1, \\ A\tilde{x}_t &= \Big[-R^{\text{data}} \quad -\mathbf{1}_q \quad -I_q \Big] \tilde{x}_t \leq b = 0_q, \\ \alpha &\geq 0, \quad u_k \geq 0 \quad \text{for } k = 1, \ldots, q. \end{aligned} \tag{A.5}$$

Here $D \in \mathbb{R}^{1 \times (N+q+1)}$ encodes the budget constraint, while $A \in \mathbb{R}^{q \times (N+q+1)}$ and $b \in \mathbb{R}^q$ encode the q inequality constraints arising from the linearization (A.4). The matrix $R^{\text{data}} \in \mathbb{R}^{q \times N}$ is comprised of sampled return vectors $y_k^\top$ row by row, 0_q is the q-dimensional zero vector, and I_q the q-dimensional identity matrix.

Problem (A.5) has a linear objective subject to one linear equality and q linear inequalities, and is thus a linear program solvable with standard methods. This formulation serves as the base unregularized portfolio optimization problem under Expected Shortfall. Regularization terms and additional constraints are added in Section A.3 of the appendix.

A.3 Regularization

The base problem (A.5) is known to be numerically unstable in high-dimensional settings, producing extreme and concentrated weights [3, 5]. We stabilize it with three regularization techniques. An L_1-norm penalty that promotes sparsity, an L_2-norm penalty that promotes shrinkage, and a leverage constraint that bounds gross exposure. Each is appended to (A.5) while preserving its linear or quadratic structure. The L_1 penalty and the leverage constraint share a common variable decomposition, which we introduce first.

A.3.1 L_1 Regularization

The L_1 penalty $\lambda_1\|x_t\|_1 = \lambda_1 \sum_{i=1}^{N} |x_{i,t}|$, with $\lambda_1 \geq 0$, is non-linear through the absolute value. Following [9], we linearize it by splitting each weight into its positive and negative parts via the auxiliary variables $\Delta_{i,t}^{+} \geq 0$ and $\Delta_{i,t}^{-} \geq 0$

$$x_{i,t} = \Delta_{i,t}^{+} - \Delta_{i,t}^{-}, \qquad |x_{i,t}| = \Delta_{i,t}^{+} + \Delta_{i,t}^{-}. \tag{A.6}$$

The second identity holds at the optimum. Minimizing the L_1 penalty drives the parts to their smallest feasible values, so at most one of $\Delta_{i,t}^{+}$, $\Delta_{i,t}^{-}$ is non-zero, giving $\Delta_{i,t}^{+} = \max(x_{i,t}, 0)$ and $\Delta_{i,t}^{-} = \max(-x_{i,t}, 0)$. The L_1 penalty is then linear in the auxiliary variables

$$\lambda_1\|x_t\|_1 = \lambda_1 \mathbf{1}_N^{\top}(\Delta_t^{+} + \Delta_t^{-}).$$

We extend the decision vector to carry the two parts

$$\tilde{x}_t = \left[x_t^{\top}, \quad \alpha, \quad u^{\top}, \quad (\Delta_t^{-})^{\top}, \quad (\Delta_t^{+})^{\top}\right]^{\top} \in \mathbb{R}^{3N+q+1},$$

and append the penalty coefficients to the cost vector

$$\tilde{c} = \left[-\mu_t^{\top}, \quad \tfrac{\gamma}{2}, \quad \tfrac{\gamma}{2q(1-\beta)}\mathbf{1}_q^{\top}, \quad \lambda_1 \mathbf{1}_N^{\top}, \quad \lambda_1 \mathbf{1}_N^{\top}\right]^{\top}.$$

The decomposition (A.6) is enforced as N additional linear equality constraints, $x_t + \Delta_t^{-} - \Delta_t^{+} = 0_N$, alongside the original budget constraint. All relations remain linear, so the optimization problem is still a linear program after the addition of the L_1 penalty.

A.3.2 L_2 Regularization

The L_2 penalty $\lambda_2\|x_t\|_2^2 = \lambda_2 x_t^{\top} x_t$, with $\lambda_2 \geq 0$, is quadratic in the weights and is written as a quadratic form

$$\lambda_2\|x_t\|_2^2 = \tfrac{1}{2}\, x_t^{\top} Q\, x_t, \qquad Q = 2\lambda_2 I_N.$$

Embedding this in the augmented vector $\tilde{x}_t$, the penalty becomes $\frac{1}{2}\,\tilde{x}_t^{\top}\tilde{Q}\,\tilde{x}_t$ with $\tilde{Q}$ placing the block $2\lambda_2 I_N$ on the asset weights and zeros on the auxiliary variables

$$\tilde{Q} = \begin{bmatrix} 2\lambda_2 I_N & 0_{N\times(2N+q+1)} \\ 0_{(2N+q+1)\times N} & 0_{(2N+q+1)\times(2N+q+1)} \end{bmatrix}.$$

Adding this term turns the L_1-augmented linear objective of the problem (A.5) into a quadratic objective $\tilde{c}^{\top}\tilde{x}_t + \frac{1}{2}\,\tilde{x}_t^{\top}\tilde{Q}\,\tilde{x}_t$. Since $\tilde{Q}$ is positive semi-definite, the objective remains convex, and the optimization problem becomes a quadratic program with linear constraints.

A.3.3 Leverage Constraint

The leverage constraint $\|x_t\|_1 \leq \ell$, with $\ell \geq 1$, bounds the portfolio's gross exposure and guarantees a bounded feasible region even when $\lambda_1 = 0$. Reusing the decomposition (A.6), it is expressed as a single linear inequality

$$\|x_t\|_1 = \mathbf{1}_N^{\top}(\Delta_t^{+} + \Delta_t^{-}) \leq \ell. \tag{A.7}$$

The choice $\ell = 1$ rules out leverage entirely and is equivalent to a hard L_1 constraint. As a linear inequality, it preserves the quadratic program structure.

A.3.4 Quadratic Program Representation of the Regularized Portfolio Optimization

Adding all three regularization terms to the base problem (A.5), makes the portfolio optimization problem under Expected Shortfall a quadratic program with linear constraints

$$\begin{aligned} \tilde{x}_t^* &= \arg\min_{\tilde{x}_t} \quad \tilde{c}^\top \tilde{x}_t + \tfrac{1}{2}\,\tilde{x}_t^\top \tilde{Q}\,\tilde{x}_t, \\ \text{s.t.} \quad \tilde{D}\tilde{x}_t &= \begin{bmatrix} \mathbf{1}_N^\top & 0_{q+1}^\top & 0_N^\top & 0_N^\top \\ I_N & 0_{N\times(q+1)} & I_N & -I_N \end{bmatrix} \tilde{x}_t = \tilde{E} = \begin{bmatrix} 1 \\ 0_N \end{bmatrix}, \\ \tilde{A}\tilde{x}_t &= \begin{bmatrix} -R^{\text{data}} & -\mathbf{1}_q & -I_q & O_{q\times N} & O_{q\times N} \\ 0_N^\top & 0 & 0_q^\top & \mathbf{1}_N^\top & \mathbf{1}_N^\top \end{bmatrix} \tilde{x}_t \le b = \begin{bmatrix} 0_q \\ \ell \end{bmatrix}, \\ &\alpha \ge 0, \quad u_k \ge 0, \quad \Delta_{i,t}^- \ge 0, \quad \Delta_{i,t}^+ \ge 0 \quad \text{for } i = 1,\dots,N,\ k = 1,\dots,q. \end{aligned} \tag{A.8}$$

The first row of $\tilde{D}$ enforces the budget constraint and the remaining N rows enforce the weight decomposition (A.6). The first q rows of $\tilde{A}$ encode the inequality constraints from the linearization (A.4), and are extended with zeros for the new variables, and the final row of $\tilde{A}$ imposes the leverage constraint (A.7).

A.4 Separate (Overlay) Optimization

In the separate (currency overlay) approach, the asset and currency allocations are optimized in two sequential steps. First, the asset weights $x_t \in \mathbb{R}^N$ are chosen on the fully hedged returns, then, conditional on the optimal x_t^*, the currency exposures $\psi_t \in \mathbb{R}^M$ are optimized. Let

$$\begin{aligned} \mu^{fh} &= \mathbb{E}[R_{t+1}^{fh}] \in \mathbb{R}^N \\ \mu^{c} &= \mathbb{E}[e_{t+1} - f_t] \in \mathbb{R}^M \end{aligned}$$

denote the expected fully hedged asset return and the expected excess currency return respectively. The two otimization steps are then given by

$$\begin{aligned} x_t^* = \arg\min_{x_t} &\left\{ -x_t^\top \mu^{fh} + \tfrac{\gamma}{2}\text{ES}_\beta(x_t^\top R_{t+1}^{fh}) + \lambda_1^a \|x_t\|_1 + \lambda_2^a \|x_t\|_2^2 \right\} \\ &\text{s.t. } x_t^\top \mathbf{1}_N = 1,\ \|x_t\|_1 \le \ell, \end{aligned} \tag{A.9}$$

$$\psi_t^* \mid x_t^* = \arg\min_{\psi_t} \left\{ -\psi_t^\top \mu^c + \tfrac{\gamma}{2}\text{ES}_\beta\Big(x_t^{*\top} R_{t+1}^{fh} + \psi_t^\top (e_{t+1} - f_t)\Big) + \lambda_1^c \|\psi_t\|_1 + \lambda_2^c \|\psi_t\|_2^2 \right\}. \tag{A.10}$$

The regularization parameters are kept separate across assets $(\lambda_1^a, \lambda_2^a)$ and currencies $(\lambda_1^c, \lambda_2^c)$.

A.4.1 Asset Optimization

The asset step (A.9) is identical to the regularized single-currency program (A.8). It carries a budget constraint, an L_1 and L_2 penalty terms, and a leverage constraint. It is therefore obtained directly from (A.8) by substituting the matrix of historically sampled fully hedged returns $R^{fh} \in \mathbb{R}^{q\times N}$ for R^{data}, the mean μ^{fh} for μ_t, and the asset-specific penalties λ_1^a, λ_2^a. With the augmented vector

$$\tilde{x}_t = [\, x_t^\top,\ \alpha_x,\ u_x^\top,\ (\Delta_{x,t}^-)^\top,\ (\Delta_{x,t}^+)^\top \,]^\top \in \mathbb{R}^{3N+q+1},$$

the cost vector and penalty matrix are

$$\tilde{c} = [\,-(\mu^{fh})^\top,\ \tfrac{\gamma}{2},\ \tfrac{\gamma}{2q(1-\beta)}\mathbf{1}_q^\top,\ \lambda_1^a\mathbf{1}_N^\top,\ \lambda_1^a\mathbf{1}_N^\top\,]^\top, \qquad \tilde{Q} = \begin{bmatrix} 2\lambda_2^a I_N & 0_{N\times(2N+q+1)} \\ 0_{(2N+q+1)\times N} & 0_{(2N+q+1)\times(2N+q+1)} \end{bmatrix},$$

and the equality, inequality, and sign constraints are exactly those of (A.8) with R^{fh} in place of R^{data}. The step yields the optimal asset weights x_t^*, which are then fixed in the currency overlay optimization.

A.4.2 Currency Overlay Optimization

Conditional on x_t^*, the currency step (A.10) minimizes the Expected Shortfall of the combined return $x_t^{*\top} R_{t+1}^{fh} + \psi_t^\top(e_{t+1} - f_t)$ over the currency exposures ψ_t. Applying the Rockafellar and Uryasev (2000) representation and the empirical approximation as in Appendix A.2, we introduce the auxiliary variable α_ψ and slacks $u_{k,\psi} \geq 0$. Since the asset leg is now fixed, the slack constraint becomes

$$u_{k,\psi} \geq -\psi_t^\top(e_k - f_t) - x_t^{*\top} R_k^{fh} - \alpha_\psi \quad \Longleftrightarrow \quad -\psi_t^\top(e_k - f_t) - \alpha_\psi - u_{k,\psi} \leq x_t^{*\top} R_k^{fh},$$

where R_k^{fh} is the k-th sampled fully hedged return vector. The fixed asset-portfolio returns therefore appear on the right-hand side. Collecting them in the vector $R_P^{fh} = R^{fh} x_t^* \in \mathbb{R}^q$ and linearizing the currency L_1 penalty with the decomposition $\psi_{c,t} = \Delta_{\psi,c,t}^+ - \Delta_{\psi,c,t}^-$ as in Appendix A.3.1, we form the augmented vector

$$\tilde{\psi}_t = [\,\psi_t^\top,\ \alpha_\psi,\ u_\psi^\top,\ (\Delta_{\psi,t}^-)^\top,\ (\Delta_{\psi,t}^+)^\top\,]^\top \in \mathbb{R}^{3M+q+1}.$$

The currency overlay is then the quadratic program

$$\begin{aligned}
\psi_t^* &= \arg\min_{\tilde{\psi}_t} \quad \tilde{c}^\top\tilde{\psi}_t + \tfrac{1}{2}\,\tilde{\psi}_t^\top\tilde{Q}\,\tilde{\psi}_t, \\
\tilde{c} &= [\,-(\mu^c)^\top,\ \tfrac{\gamma}{2},\ \tfrac{\gamma}{2q(1-\beta)}\mathbf{1}_q^\top,\ \lambda_1^c\mathbf{1}_M^\top,\ \lambda_1^c\mathbf{1}_M^\top\,]^\top, \\
\tilde{Q} &= \begin{bmatrix} 2\lambda_2^c I_M & 0_{M\times(2M+q+1)} \\ 0_{(2M+q+1)\times M} & 0_{(2M+q+1)\times(2M+q+1)} \end{bmatrix}, \\
\text{s.t.} \quad \tilde{D}\tilde{\psi}_t &= \begin{bmatrix} I_M & 0_{M\times(q+1)} & I_M & -I_M \end{bmatrix}\tilde{\psi}_t = \tilde{E} = 0_M, \\
\tilde{A}\tilde{\psi}_t &= \begin{bmatrix} -R^c & -\mathbf{1}_q & -I_q & O_{q\times M} & O_{q\times M} \end{bmatrix}\tilde{\psi}_t \leq b = R_P^{fh} = R^{fh}x_t^*, \\
\alpha_\psi \geq 0, \quad u_{j,\psi} \geq 0, \quad \Delta_{\psi,c,t}^- \geq 0, \quad \Delta_{\psi,c,t}^+ \geq 0 \quad &\text{for } c = 1,\dots,M,\ j = 1,\dots,q.
\end{aligned} \tag{A.11}$$

The currency optimization differs from the asset optimization (A.8) in the following ways. First, there is no budget constraint, the currency portfolio is a zero-investment portfolio, so the domestic exposure $\psi_{0,t}$ is inferred residually and the equality block $\tilde{D}$ enforces only the weight decomposition, with right-hand side $\tilde{E} = 0_M$ rather than $[1, 0_M^\top]^\top$. Second, the fixed asset leg shifts the inequality right-hand side from 0_q to $R_P^{fh} = R^{fh}x_t^*$. Third, no leverage constraint is imposed on the currency exposures, so $\tilde{A}$ contains only the q Rockafellar and Uryasev (2000) rows. The L_1 and L_2 regularizing terms are as before, through the last two blocks of $\tilde{c}$ and the main block of $\tilde{Q}$, so the currency overlay remains a convex quadratic program.

A.5 Joint Optimization

In the joint approach, the asset and currency decisions are made simultaneously rather than sequentially. We stack them into the combined decision vector $\theta_t = (x_t, \psi_t)^\top \in \mathbb{R}^{N+M}$ and the

combined return vector $R_{t+1} = (R^{fh}_{t+1}, e_{t+1} - f_t)^\top \in \mathbb{R}^{N+M}$, with joint mean $\mu = \mathbb{E}[R_{t+1}] \in \mathbb{R}^{N+M}$. The joint problem is

$$\begin{aligned} \theta_t^* = \arg\min_{\theta_t} \Big\{ -\theta_t^\top \mu + \tfrac{\gamma}{2} \mathrm{ES}_\beta(\theta_t^\top R_{t+1}) + \|S_1\theta_t\|_1 + \|S_2\theta_t\|_2^2 \Big\}, \\ \text{s.t. } \theta_t^\top q_{N,M} = 1, \quad \|L\theta_t\|_1 \le \ell. \end{aligned} \tag{A.12}$$

The combined return satisfies $\theta_t^\top R_{t+1} = x_t^\top R^{fh}_{t+1} + \psi_t^\top (e_{t+1} - f_t)$, so a single Expected Shortfall term now depends jointly on all weights, removing the two-step structure of the separate approach.

Four structured objects adapt the single-currency machinery to the joint setting, all respecting the asymmetry between assets and currencies:

$$q_{N,M} = \begin{bmatrix} \mathbf{1}_N \\ 0_M \end{bmatrix}, \quad L = \begin{bmatrix} I_N & 0_{N\times M} \\ 0_{M\times N} & 0_{M\times M} \end{bmatrix}, \quad S_1 = \begin{bmatrix} \lambda_1^a I_N & 0 \\ 0 & \lambda_1^c I_M \end{bmatrix}, \quad S_2 = \begin{bmatrix} \sqrt{\lambda_2^a}\, I_N & 0 \\ 0 & \sqrt{\lambda_2^c}\, I_M \end{bmatrix}.$$

The selector $q_{N,M}$ applies the budget constraint to the asset block only, leaving the currency exposures as a zero-investment hedge; the diagonal matrix L likewise applies the leverage constraint to the assets only, so that $\|L\theta_t\|_1 = \|x_t\|_1$. The diagonal regularization matrices S_1 and S_2 let the asset and currency penalties differ within one objective,

$$\|S_1\theta_t\|_1 = \lambda_1^a \|x_t\|_1 + \lambda_1^c \|\psi_t\|_1, \qquad \|S_2\theta_t\|_2^2 = \lambda_2^a \|x_t\|_2^2 + \lambda_2^c \|\psi_t\|_2^2,$$

where the square roots in S_2 ensure the squared norm reproduces the intended coefficients λ_2^a, λ_2^c.

We apply the Rockafellar and Uryasev (2000) representation and the empirical approximation of Appendix A.2 once to the joint return $\theta_t^\top R_{t+1}$, introducing the auxiliary variable α and slacks $u_k \ge 0$, and linearize the L_1 penalty by decomposing all $N+M$ components of θ_t into positive and negative parts as in Appendix A.3.1. Collecting the asset and currency parts separately, we form the augmented vector

$$\tilde\theta_t = [\, x_t^\top,\ \psi_t^\top,\ \alpha,\ u^\top,\ (\Delta^-_{x,t})^\top,\ (\Delta^-_{\psi,t})^\top,\ (\Delta^+_{x,t})^\top,\ (\Delta^+_{\psi,t})^\top \,]^\top \in \mathbb{R}^{3N+3M+q+1}.$$

With $R^{\text{all}} = [\, R^{fh} \ \ R^c \,] \in \mathbb{R}^{q\times(N+M)}$ the matrix of jointly sampled asset and currency returns, the joint problem is the quadratic program

$$\begin{aligned} \tilde\theta_t^* &= \arg\min_{\tilde\theta_t} \quad \tilde c^\top \tilde\theta_t + \tfrac{1}{2}\, \tilde\theta_t^\top \tilde Q\, \tilde\theta_t, \\ \tilde c &= [\, -\mu^\top,\ \tfrac{\gamma}{2},\ \tfrac{\gamma}{2q(1-\beta)} \mathbf{1}_q^\top,\ \lambda_1^a \mathbf{1}_N^\top,\ \lambda_1^c \mathbf{1}_M^\top,\ \lambda_1^a \mathbf{1}_N^\top,\ \lambda_1^c \mathbf{1}_M^\top \,]^\top, \\ \tilde Q &= \begin{bmatrix} 2\lambda_2^a I_N & 0 & 0 \\ 0 & 2\lambda_2^c I_M & 0 \\ 0 & 0 & 0_{(2N+2M+q+1)\times(2N+2M+q+1)} \end{bmatrix}, \\ \text{s.t.} \quad \tilde D\tilde\theta_t &= \begin{bmatrix} q_{N,M}^\top & 0_{q+1}^\top & 0_{N+M}^\top & 0_{N+M}^\top \\ I_{N+M} & 0_{(N+M)\times(q+1)} & I_{N+M} & -I_{N+M} \end{bmatrix} \tilde\theta_t = \tilde E = \begin{bmatrix} 1 \\ 0_{N+M} \end{bmatrix}, \\ \tilde A\tilde\theta_t &= \begin{bmatrix} -R^{\text{all}} & -\mathbf{1}_q & -I_q & O_{q\times(N+M)} & O_{q\times(N+M)} \\ 0_{N+M}^\top & 0_{q+1}^\top & q_{N,M}^\top & q_{N,M}^\top & \end{bmatrix} \tilde\theta_t \le b = \begin{bmatrix} 0_q \\ \ell \end{bmatrix}, \\ \alpha \ge 0, \quad u_j \ge 0, \quad \Delta^\pm_{x,i,t} \ge 0, \quad \Delta^\pm_{\psi,c,t} \ge 0 \quad &\text{for } i = 1,\dots,N,\ c = 1,\dots,M,\ j = 1,\dots,q. \end{aligned} \tag{A.13}$$

The first row of $\tilde{D}$ imposes the asset budget through the selector $q_{N,M}^{\top} = [\mathbf{1}_N^{\top},\ 0_M^{\top}]$, and the remaining $N + M$ rows enforce the weight decomposition for the combined vector. In $\tilde{A}$, the first q rows are the Rockafellar and Uryasev (2000) inequalities on the joint return matrix R^{all}, with right-hand side 0_q — in contrast to the conditional currency step (A.11), no asset leg is pre-fixed, so nothing shifts to the right-hand side. The final row imposes the asset-only leverage constraint, where $q_{N,M}^{\top}$ again selects the asset parts of the decomposition, giving $\mathbf{1}_N^{\top}(\Delta_{x,t}^{+} + \Delta_{x,t}^{-}) = \|x_t\|_1 \leq \ell$.

The joint formulation thus solves a single quadratic program over the combined vector, capturing the dependencies between asset and currency exposures directly through the joint Expected Shortfall term and the off-diagonal return covariation embedded in R^{all}. This comes at the cost of a higher-dimensional problem ($3N + 3M + q + 1$ variables) relative to the two smaller programs of the separate approach, which is the trade-off examined empirically in the main text.

Appendix B

Optimization Derivation: Mean-Variance

Appendix C

Transaction Costs

Following the methodology outlined by [11], [11] and [**?**], and keeping the notation introduced in Chapter **??**, we model transaction costs as proportional to the volume of traded assets and currencies, ignoring any fixed trading costs. The return of the hedged portfolio is expressed as the sum of unhedged asset returns and gains from forward currency positions

$$R^h_{P,t+1} = \sum_{i=1}^{N} x_{i,t} R^u_{i,t+1} + \sum_{c=1}^{M} \varphi_{c,t}(f_{c,t} - e_{c,t+1}). \tag{(3.2)}$$

Let τ^{asset}_{t+1} and τ^{curr}_{t+1} denote the total transaction costs associated with asset and currency trades, respectively. The portfolio return net of transaction costs is then given by

$$R^{h,net}_{P,t+1} = R^h_{P,t+1} - \tau^{asset}_{t+1} - \tau^{curr}_{t+1}. \tag{C.1}$$

The total transaction costs are computed as the sum of the costs incurred from trading individual assets and currencies, weighted by the respective trading volumes

$$\tau^{asset}_{t+1} = \sum_{i=1}^{N} \tau^{asset}_{i,t+1} |x_{i,t+1} - x^-_{i,t+1}|,$$

$$\tau^{curr}_{t+1} = \sum_{c=1}^{M} \tau^{curr}_{c,t+1} \varphi_{c,t+1} + \sum_{c=1}^{M} \tau^{curr}_{c,t+1} |w_{c,t+1} - w^-_{c,t+1}|,$$

where $\tau^{asset}_{i,t+1}$ and $\tau^{curr}_{c,t+1}$ are the per-unit transaction costs for asset i and currency c. The terms $x^-_{i,t+1}$ and $w^-_{c,t+1}$ denote the asset weights and implied currency weights immediately prior to rebalancing at time $t+1$ (i.e., the drifted weights).

Asset transaction costs are incurred exclusively when rebalancing individual asset holdings. Currency transaction costs, however, arise from two sources, adjustments in forward hedge positions (assuming the forward contract maturity matches the rebalancing frequency, the old contract expires and a new contract of full notional size $\varphi_{c,t+1}$ must be established at each rebalancing date) and currency exchanges required to fund purchases of assets denominated in different currencies ($|w_{c,t+1} - w^-_{c,t+1}|$). If the portfolio is rebalanced more frequently than the forward contract maturity, existing forward contracts are kept open and only incremental adjustments are traded. In this case, the forward transaction cost term is modeled as the absolute difference, replacing $\sum_{c=1}^{M} \tau^{curr}_{c,t+1} \varphi_{c,t+1}$ with $\sum_{c=1}^{M} \tau^{curr}_{c,t+1} |\varphi_{c,t+1} - \varphi^-_{c,t+1}|$, where $\varphi^-_{c,t+1}$ represents the drifted hedge weight. Following [11], these costs are aggregated at the currency level, operating under the assumption that proceeds from selling assets denominated in

currency c can fund the purchase of other assets in the same currency without incurring currency conversion fees.

To define the individual per-unit costs, we consider an asset i denominated in currency c_i. Let $P^{ask}_{i,t}$, $P^{bid}_{i,t}$, and $P_{i,t}$ represent the ask, bid, and mid prices of the asset, while $S^{ask}_{c_i,t}$, $S^{bid}_{c_i,t}$, and $S_{c_i,t}$ denote the corresponding currency prices. If asset i is purchased at time t and sold at time $t+1$, its realized log return is given by

$$\begin{aligned}
\log\left(\frac{P^{bid}_{i,t+1}S^{bid}_{c_i,t+1}}{P^{ask}_{i,t}S^{ask}_{c_i,t}}\right) &= \log\left(\frac{\left(P_{i,t+1}-\frac{\delta^{asset}_{i,t+1}}{2}\right)\left(S_{c_i,t+1}-\frac{\delta^{curr}_{c_i,t+1}}{2}\right)}{\left(P_{i,t}+\frac{\delta^{asset}_{i,t}}{2}\right)\left(S_{c_i,t}+\frac{\delta^{curr}_{c_i,t}}{2}\right)}\right) \\
&= \log\left(\frac{P_{i,t+1}\left(1-\frac{\delta^{asset}_{i,t+1}}{2P_{i,t+1}}\right)S_{c_i,t+1}\left(1-\frac{\delta^{curr}_{c_i,t+1}}{2S_{c_i,t+1}}\right)}{P_{i,t}\left(1+\frac{\delta^{asset}_{i,t}}{2P_{i,t}}\right)S_{c_i,t}\left(1+\frac{\delta^{curr}_{c_i,t}}{2S_{c_i,t}}\right)}\right) \\
&= \log\left(\frac{P_{i,t+1}S_{c_i,t+1}}{P_{i,t}S_{c_i,t}}\right)+\log\left(\frac{1-\frac{\delta^{asset}_{i,t+1}}{2P_{i,t+1}}}{1+\frac{\delta^{asset}_{i,t}}{2P_{i,t}}}\right)+\log\left(\frac{1-\frac{\delta^{curr}_{c_i,t+1}}{2S_{c_i,t+1}}}{1+\frac{\delta^{curr}_{c_i,t}}{2S_{c_i,t}}}\right) \\
&= \log\left(\frac{P_{i,t+1}S_{c_i,t+1}}{P_{i,t}S_{c_i,t}}\right)-\log\left(\frac{1+\frac{\delta^{asset}_{i,t}}{2P_{i,t}}}{1-\frac{\delta^{asset}_{i,t+1}}{2P_{i,t+1}}}\right)-\log\left(\frac{1+\frac{\delta^{curr}_{c_i,t}}{2S_{c_i,t}}}{1-\frac{\delta^{curr}_{c_i,t+1}}{2S_{c_i,t+1}}}\right),
\end{aligned}$$

where δ^{asset}_i and $\delta^{curr}_{c_i}$ are the bid-ask spreads. Comparing this to equation (C.1), the transaction cost is precisely the difference between the actual realized log return and mid-price returs for the asset and currency.

Defining half of the proportional bid-ask spreads as

$$\chi^{asset}_{i,t}=\frac{P^{ask}_{i,t}-P^{bid}_{i,t}}{2P_{i,t}} \quad \text{and} \quad \chi^{curr}_{c,t}=\frac{S^{ask}_{c,t}-S^{bid}_{c,t}}{2S_{c,t}},$$

the the per-unit transaction costs for trading asset i and currency c become

$$\tau^{asset}_{i,t+1}=\begin{cases}\log\left(\frac{1+\chi^{asset}_{i,t+1}}{1-\chi^{asset}_{i,t}}\right) & \text{if } x_{i,t+1}>x^{-}_{i,t+1},\\ \log\left(\frac{1-\chi^{asset}_{i,t}}{1+\chi^{asset}_{i,t+1}}\right) & \text{otherwise,}\end{cases}$$

and

$$\tau^{curr}_{c,t+1}=\begin{cases}\log\left(\frac{1+\chi^{curr}_{c,t+1}}{1-\chi^{curr}_{c,t}}\right) & \text{if } \varphi_{c,t+1}>0 \text{ or } w_{c,t+1}>w^{-}_{c,t+1},\\ \log\left(\frac{1-\chi^{curr}_{c,t}}{1+\chi^{curr}_{c,t+1}}\right) & \text{otherwise.}\end{cases}$$

For our empirical analysis, we adopt the simplifying assumption, that proportional bid-ask spreads remain constant both over time and across the cross-sections of both assets and currencies. We set the half-spreads to $\chi^{asset}=0.001$ (10 bps) for all equities and $\chi^{curr}=0.0001$ (1 bp) for all currencies, consistent with empirical finidngs in [**?**] and [**?**]. Under these assumptions, the transaction costs simplify to constant values of $\tau^{asset}\approx 0.002$ and $\tau^{curr}\approx 0.0002$.

Appendix D

Hyperparameter Tuning and Additional Results

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